Q1. What is java and explain the features of JAVA?
Q2. Why java is a pure object oriented language explain any 5 reasons?
Q3. Why java is not pure object oriented explain any 2 reasons?
Q4. What is JDK , JRE and JVM and explain their differences?
Q5. What is compiling and explaining their roles?
Q6. What is JVM and explain their role?
Q7. What is byte code and what is the benefit of byte code?
Q8. What is a class file and when is it created?
Q9. Why is java platform independent?
Q10.What is platform independence?
Q11.Why is the main method static?
Q12.Can we overload the main method in Java?
Q13.Can we override the main method in Java?
Q14.What is the command line argument and why use it?
Q15.What is the Scanner class and why use it?
Q16.What is data type and explain all types of data type?
Q17.what is constant and literals in Java?
Q18.What is the difference between primitive and referential data type?
Q19.What is object oriented concept?
Q20.What is the meaning of cross platform?
Q21.explain all types of operators in java?
Q22.explain all bitwise operators?
Q23 explain conditional operators in java?
Q24.Explain loop and why use it?
Q25.What is break and continue?
Q26.Explain types of loop?
Q27.What is the difference between break and continue keyword?
Q28.What is array in java and why use array?
Q29.Explain how many ways to declare arrays in java?
Q30.Explain types of arrays in Java?
Q31.What is an anonymous array in Java?
Q32.How to initialize an array in Java?

Time complexity related question?

Q1. What is time complexity and why use it?
Q2. Explain how we can calculate process operation?
Q3. Explain constant time complexity i.e O(1)?
Q4. Explain constant linear complexity i.e O(N)?
Q5. Explain quadratic time complexity i.e O(n^2)
Q6. Explain logarithmic complexity like O(log n) or O(n log n)?
Q7. $O(2^n)$ – Exponential Time ?
Q8. O(n!) – Factorial Time ?
Q9. Explain priority of time complexity?
Q10. Explain the space complexity and how we can calculate it?
Q11. What is the best case, average case or worst case time complexity?
Q12. What is constant space complexity?
Q13. What is linear space complexity?
Q14. What is quadratic space complexity?
Q15. What is auxiliary space?

Interview question on function and recursion

Q1.How would you define a function?
Q2.What is the difference between a function and a method?
Q3.Can you explain the difference between function declaration and function definition?
Q4.What is the difference between call by value and call by reference?
Q5.What are pure functions? Can you give an example?
Q6.What do you mean by side effects in functions?
Q7.What is function overloading? How is it implemented in Java/C++?
Q8.What is recursion? How does it work?
Q9.What are base cases and recursive cases in recursion?
Q10.How does recursion work internally in memory (call stack)?
Q11.What is the difference between recursion and iteration?
Q12.What are the advantages and disadvantages of recursion?
Q13.What is the difference between tail recursion and non-tail recursion?
Q14. What is stack overflow in recursion and why does it occur?
Q15.Can you explain the recursion tree concept?
Q16.What is the difference between backtracking and recursion?
Q17.What is memoization? How does it optimize recursive solutions?
OOP

Interview Question on function overloading and polymorphism

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1. Function Overloading Interview Questions
    What is function overloading?
    How does function overloading work in Java/C++?
    What are the rules of function overloading?
    Can we overload functions based only on return type? Why or why not?
    What is compile-time polymorphism? How is it related to overloading?
    How does the compiler differentiate between overloaded methods?
    Can we overload the main() method?
    What is method signature in overloading?
    Difference between method overloading and method overriding

2. Polymorphism Interview Questions
    What is polymorphism in OOP?
    What are the types of polymorphism?
    What is compile-time polymorphism?
    What is runtime polymorphism?
    What is method overriding?
    What are the rules of method overriding?
    What is dynamic method dispatch?
    How does Java achieve runtime polymorphism?

3. Conceptual / Deep Understanding Questions
    Difference between overloading and overriding
    Why is polymorphism important in OOP?

Can constructors be overloaded?
Can static methods be overridden? Why or why not?
What is early binding vs late binding?
What is the role of inheritance in polymorphism?
Can private methods be overridden?

4. Frequently Asked Questions
What happens if method signatures are same but return types differ?
Can we overload methods with different access modifiers?
What is covariant return type?
What is the difference between compile-time error and runtime behavior in polymorphism?
Why method overloading is faster than overriding?

Interview Question on classes and objects

Q1. What is class?
Q2. Why use class or Benefits of class?
Q3. How to reuse class more than one time?
Q4. What is an object and how to create an object in Java?
Q5. What is the difference between reference and object?
Q6. How to pass parameters to call function?
Q7. How to pass an array as a parameter to class function?
Q8. What is a method with variable argument concepts?
Q9. What is POJO class & Why use POJO class?
Q10. Static variable & static methods ?
Q11. Local Variable , instance variable?
Q12. What is the difference between local variable ,instance variable and static variable?
Q13. Array of objects?
Q14. What is encapsulation & Benefit of Encapsulation?
Q15. What are nested classes and why use nested class?
Q16. What is a simple nested class?
Q17. What is a static nested class?
Q18. What is the local nested class?
Q19. What are anonymous nested classes?

Interview question on constructor

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1. Basic Constructor Questions
What is a constructor?
What is the purpose of a constructor?
How is a constructor different from a method?
What are the rules of constructors?
When is a constructor called?
Can a constructor return a value?

2. Types of Constructors

What is a default constructor?
What is a parameterized constructor?
What is a copy constructor (in C++)?
What is a no-argument constructor?
Difference between default and parameterized constructor

3. Constructor Overloading

What is constructor overloading?
Can constructors be overloaded? How?
Difference between constructor overloading and method overloading
How does constructor overloading support polymorphism?

4. Constructor Behavior & Rules

What happens if no constructor is defined?
Can we make a constructor private? Why?
Can constructors be static? Why or why not?
Can constructors be final or abstract?
Can we inherit constructors?

5. Constructor Chaining

What is constructor chaining?
What is the use of this()?
What is the use of super()?
Difference between this() and super()
Can we call both this() and super() in same constructor?

6. Advanced / Conceptual Questions

What is the order of constructor execution in inheritance?
What is object initialization?
Can a constructor call a method?
What happens if constructor throws an exception?
What is the difference between constructor and initialization block?

7. Tricky / Frequently Asked Questions

Can we override a constructor?
Why constructors cannot be inherited?
Can an interface have a constructor?
Can abstract class have a constructor?
What is the difference between constructor and destructor (C++)?

9. Interview Focus (Most Important)

Constructor vs Method
Constructor overloading
this() vs super()

Constructor chaining
Rules (no return type, same name as class)

MCQ Question

Q1. what will be output of given code?

```
class Q1 {
    int x = 5;
    public static void main(String[] args) {
        Q1 obj = new Q1();
        System.out.println(obj.x);
    }
}
```

Q2. what will be output of given code?

```
class Q2 {
    int a;
    Q2() { a = 10; }
    public static void main(String[] args) {
        Q2 obj = new Q2();
        System.out.println(obj.a);
    }
}
```

Q3. what will be output of given code?

```
class Q3 {
    int x = 1;
    int y = x + 1;
    public static void main(String[] args) {
        Q3 obj = new Q3();
        System.out.println(obj.y);
    }
}
```

Q4.what will be output of given code?

```
class Q4 {
    int a = 10;
    Q4(int a) { this.a = a; }
    public static void main(String[] args) {
        Q4 obj = new Q4(20);
        System.out.println(obj.a);
    }
}
```

Q5.what will be output of given code?

```
class Q5 {
    int x;
    void setX(int x) { x = x; }
    public static void main(String[] args) {
        Q5 obj = new Q5();
        obj.setX(10);
        System.out.println(obj.x);
    }
}
```

Q6.what will be output of given code?

```
class Q6 {
    int x = 5;
    void change(Q6 obj) { obj.x = 10; }
    public static void main(String[] args) {
        Q6 obj = new Q6();
        obj.change(obj);
        System.out.println(obj.x);
    }
}
```

Q7.what will be output of given code?

```
class Q7 {
    int a = 1;
    int b = 2;
    public static void main(String[] args) {
        Q7 obj = new Q7();
        System.out.println(obj.a + obj.b);
    }
}
```

Q8.what will be output of given code?

```
class Q8 {
    int a;
    Q8(int x) { a = x; }
    Q8() { this(5); }
    public static void main(String[] args) {
        Q8 obj = new Q8();
        System.out.println(obj.a);
    }
}
```

Q9.what will be output of given code?

```
class Q9 {
    int a = 10;
    int b = a++;
    public static void main(String[] args) {
        Q9 obj = new Q9();
        System.out.println(obj.b);
    }
```

```
}
```

Q10.what will be output of given code?

```
class Q10 {
    int a = 10;
    void test(Q10 obj) { obj.a += 5; }
    public static void main(String[] args) {
        Q10 obj = new Q10();
        obj.test(obj);
        System.out.println(obj.a);
    }
}
```

Q11.what will be output of given code?

```
class Outer {
    int x = 10;
    class Inner {
        void display() { System.out.println(x); }
    }
    public static void main(String[] args) {
        Outer o = new Outer();
        Inner i = o.new Inner();
        i.display();
    }
}
```

Q12.what will be output of given code?

```
class Outer {
    static int x = 5;
    static class Inner {
        void show() { System.out.println(x); }
    }
    public static void main(String[] args) {
        Inner i = new Inner();
        i.show();
    }
}
```

Q13.what will be output of given code?

```
class Outer {
    int a = 1;
    void outerMethod() {
        class Local {
            void display() { System.out.println(a); }
        }
        Local l = new Local();
        l.display();
    }
    public static void main(String[] args) {
        Outer o = new Outer();
        o.outerMethod();
```

```
    }
}
```

Q14.what will be output of given code?

```
class Outer {
    void display() {
        Runnable r = new Runnable() {
            public void run() { System.out.println("Anonymous"); }
        };
        r.run();
    }
    public static void main(String[] args) {
        new Outer().display();
    }
}
```

Q15.what will be output of given code?

```
class Outer {
    int x = 10;
    static class Inner {
        static int y = 20;
        void show() { System.out.println(y); }
    }
    public static void main(String[] args) {
        Inner i = new Inner();
        i.show();
    }
}
```

Q16.what will be output of given code?

```
class Outer {
    static int x = 10;
    class Inner {
        void show() { System.out.println(x); }
    }
    public static void main(String[] args) {
        Outer o = new Outer();
        Inner i = o.new Inner();
        i.show();
    }
}
```

Q17.what will be output of given code?

```
class Outer {
    int x = 5;
    class Inner {
        int x = 10;
        void show() { System.out.println(this.x); }
    }
    public static void main(String[] args) {
        Outer o = new Outer();
        Inner i = o.new Inner();
        i.show();
```

```
    }
}
```

Q18.what will be output of given code?

```
class Outer {
    int a = 1;
    static int b = 2;
    static class Inner {
        void display() { System.out.println(b); }
    }
    public static void main(String[] args) {
        Inner i = new Inner();
        i.display();
    }
}
```

Q19.what will be output of given code?

```
class Outer {
    int a = 5;
    class Inner {
        int a = 10;
        void print() { System.out.println(Outer.this.a); }
    }
    public static void main(String[] args) {
        Outer o = new Outer();
        Inner i = o.new Inner();
        i.print();
    }
}
```

Q20.what will be output of given code?

```
class Outer {
    void method() {
        final int x = 50;
        class Local {
            void show() { System.out.println(x); }
        }
        new Local().show();
    }
    public static void main(String[] args) {
        new Outer().method();
    }
}
```

Q21.what will be output of given code?

```
class Q21 {
    static int x = 10;
    public static void main(String[] args) {
        Q21.x += 5;
        System.out.println(Q21.x);
    }
```

```
}
```

Q22.what will be output of given code?

```
class Q22 {
    final int a = 10;
    void change() { a = 20; }
    public static void main(String[] args) {
        new Q22().change();
    }
}
```

Q23.what will be output of given code?

```
class Q23 {
    static int x = 5;
    static void increment() { x++; }
    public static void main(String[] args) {
        increment();
        System.out.println(x);
    }
}
```

Q24.what will be output of given code?

```
class Q24 {
    static final int a = 10;
    public static void main(String[] args) {
        System.out.println(a);
    }
}
```

Q25.what will be output of given code?

```
class Q25 {
    final void show() { System.out.println("Final Method"); }
    class Sub extends Q25 {
        void show() { System.out.println("Override"); }
    }
    public static void main(String[] args) {
        new Sub().show();
    }
}
```

Q26. what will be output of given code?

```
abstract class Q26 {
    abstract void show();
    public static void main(String[] args) {
        // Q26 obj = new Q26(); // Uncomment to test
    }
}
```

Q27. What will be output of given code?

```
abstract class Q27 {
    abstract void display();
```

```
}
class SubQ27 extends Q27 {
    void display() { System.out.println("Hello"); }
    public static void main(String[] args) {
        new SubQ27().display();
    }
}
```

Q28.What will be output of given code?

```
interface I28 {
    void show();
}
class Q28 implements I28 {
    public void show() { System.out.println("Interface"); }
    public static void main(String[] args) {
        new Q28().show();
    }
}
```

Q29. What will be output of given code?

```
interface I29 {
    int x = 10;
}
class Q29 implements I29 {
    void print() { System.out.println(x); }
    public static void main(String[] args) {
        new Q29().print();
    }
}
```

Q30.What will be output of given code?

```
abstract class Q30 {
    abstract void display();
    void hello() { System.out.println("Hello"); }
}
class SubQ30 extends Q30 {
    void display() { System.out.println("World"); }
    public static void main(String[] args) {
        SubQ30 obj = new SubQ30();
        obj.display();
        obj.hello();
    }
}
```